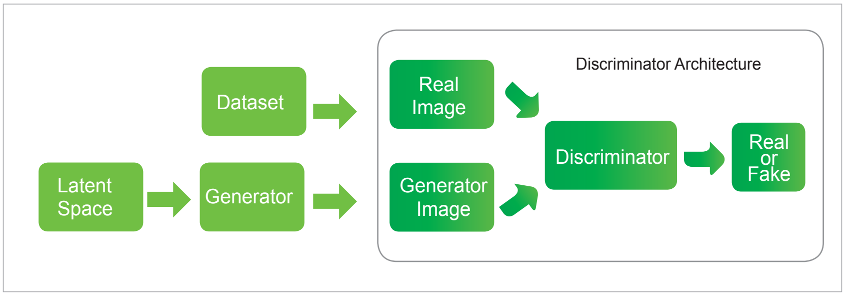

Figure 3: GAN generator

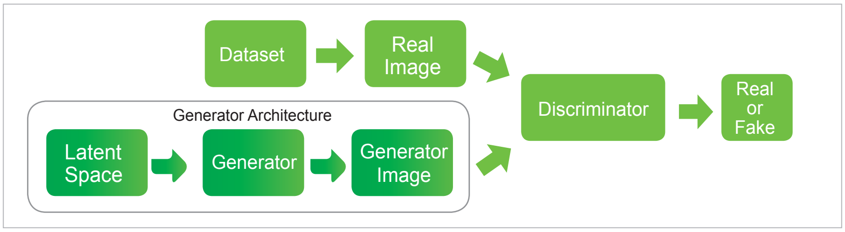

Figure 4: GAN discriminator

Magenta: A Google Brain project

Magenta was an open source project developed by Google Brain in 2016, and is aimed at creating a new tool for artists when they work on developing new songs or artwork. Magenta is powered by Google's TensorFlow machine learning platform, and can work with music and images. In the music domain, an agent automatically composes background music in real-time using the emotional state of the environment in which it is embedded. It chooses an appropriate composition algorithm dynamically from a database of previously chosen algorithms mapped to a given emotional state. Doug Eck and team worked with an LSTM model tuned with reinforcement learning. Reinforcement learning was used to teach the model to use certain rules while still allowing it to retain information learnt from data.

The LSTM model works on two metrics – the metrics we want to be low and the metrics we want to be high. The metrics associated with penalties are: a) notes not in key; b) mean autocorrelation – since the goal is to encourage variety, the model is penalised if the composition is highly correlated with itself; c) notes excessively related – LSTM is prone to repeat patterns. Reinforcement learning is brought in for creativity. The metrics associated with rewards are: a) compositions starting with a tonic note; b) leaps resolved – in order to avoid awkward intervals, leaps are taken in opposite directions, and leaping twice in the same direction is negatively rewarded; c) compositions (with unique maximum note and unique minimum note) in motif, which are a succession of notes representing a short musical idea. These metrics form a music theory rule. The degree of improvement of these metrics is determined by the reward given to a particular behaviour. The choice of metrics and the weights determine the shape of the music created. The most recent model of Magenta has used GAN and transformers to generate music with an improved long-term structure.

Challenges in music generation

The greatest challenge in the generation of music is to be able to encode various musical features. Once that is accomplished, the generative music is supposed to follow the broader structure, dynamics and rules of music. Musical dimensions such as timing and pitch have relative rather than absolute significance when it comes to how notes are placed in them. Features such as dynamics (that tells the volume of the sound from the instrument) and timbre (that differentiates between notes having the same pitch and loudness) are difficult to encode. There are other features such as duration, rest, timing and pitch that are also challenging to represent as extracted features.

Future trends

Various GAN models such as MidiNet, SSMGAN, C-RNN-GAN, JazzGAN, MuseGAN, Conditional LSTM GAN, etc, have been attempted for composing melody. Other generative models like the VAE, flow-based models, autoregressive models, transformers, RBMs, HMMs and many others have been used in research in this area. Melody generation is moving towards bringing in more musical diversity and structure handling capacity. Better interpretability and human control are being aimed for. Standardised test data sets and evaluation metrics, cross-modal generation, composition style transfer, lyrics-free singing and interactive music generation are some future research directions in this field. END

References

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